

YOKKAICHI, Japan

WMO: 476840

Lat: 34.9400N

Lon: 136.5800E

Elev: 187

StdP: 14.60

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.4	30.2	16.4	12.7	39.0	19.1	14.4	39.3	18.2	43.3	16.1	43.1	4.0	250	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	92.0	78.0	89.7	77.5	87.5	77.0	80.3	87.0	79.4	85.6	78.5	84.4	6.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.7	150.2	83.3	77.8	145.4	82.5	76.9	141.1	81.7	44.1	87.0	43.1	86.0	42.2	84.7	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.1	12.6	11.0	DB	25.0	96.0	1.6	2.4	23.9	97.7	23.0	99.1	22.1	100.5	21.0	102.2
			WB	23.2	82.2	1.6	1.4	22.1	83.2	21.1	84.0	20.3	84.7	19.1	85.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.0	40.1	41.2	46.9	56.3	64.8	71.4	79.1	81.4	74.7	64.4	53.7	44.4	(6)
	DBStd	14.96	3.90	4.58	5.22	5.61	4.15	3.99	4.06	3.34	4.80	4.98	5.31	4.81	(7)
	HDD50	901	309	248	126	12	0	0	0	0	0	0	23	183	(8)
	HDD65	3371	773	666	561	265	54	1	0	0	1	71	340	639	(9)
	CDD50	4542	0	3	30	200	460	643	902	974	740	447	133	10	(10)
	CDD65	1535	0	0	0	3	49	194	437	509	291	52	0	0	(11)
	CDH74	12233	0	0	0	24	243	924	3826	5135	1912	169	0	0	(12)
	CDH80	3853	0	0	0	1	29	171	1296	1891	454	11	0	0	(13)
(14) Wind	WSAvg	4.6	5.1	5.4	5.6	5.2	4.8	4.1	4.0	4.3	4.3	4.0	4.0	4.5	(14)
(15) Precipitation	PrecAvg	66.7	2.2	2.7	4.3	5.4	6.9	8.5	7.3	5.6	10.9	6.2	3.7	2.2	(15)
	PrecMax	86.1	6.8	5.5	8.5	9.8	15.2	12.7	14.5	16.6	22.0	18.7	11.5	4.8	(16)
	PrecMin	39.6	0.4	1.1	1.7	1.4	3.2	3.0	1.9	1.1	1.7	0.9	0.3	0.2	(17)
	PrecStd	11.7	1.6	1.2	1.7	2.4	3.0	2.7	3.7	3.7	5.5	3.5	2.7	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	60.1	66.9	77.4	83.7	88.5	95.1	96.0	90.5	81.5	71.2	61.6	(19)
		MCWB	46.7	49.6	53.5	59.5	66.0	74.8	77.7	78.6	76.8	70.2	60.7	53.7	(20)
	2%	DB	52.5	55.5	63.0	72.4	79.2	84.4	91.6	92.8	87.2	78.2	68.0	58.2	(21)
		MCWB	45.0	47.0	51.6	59.0	64.5	72.2	78.2	78.1	75.3	67.2	58.6	50.5	(22)
	5%	DB	50.1	52.8	59.7	69.3	76.6	81.5	89.1	90.5	84.8	75.5	65.4	55.6	(23)
		MCWB	43.0	45.1	50.3	57.8	63.4	70.9	77.7	77.9	74.4	66.4	57.2	48.5	(24)
	10%	DB	47.9	50.3	56.9	66.5	73.9	79.1	86.8	88.5	82.5	73.1	62.9	53.2	(25)
		MCWB	41.1	43.5	49.0	56.6	62.5	69.8	77.1	77.6	73.3	64.9	56.0	46.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	53.5	58.2	66.6	71.2	78.1	81.6	81.9	79.5	74.5	64.0	56.7	(27)
		MCDB	52.4	57.2	61.3	69.8	75.8	84.3	89.1	89.4	85.4	76.8	66.7	58.3	(28)
	2%	WB	46.3	49.7	55.5	63.4	68.6	75.8	80.0	80.8	78.1	71.6	61.5	52.5	(29)
		MCDB	50.7	53.2	59.4	67.6	73.6	80.9	86.8	87.5	82.8	75.3	65.0	55.9	(30)
	5%	WB	43.8	46.7	53.0	61.1	67.0	74.0	79.0	79.9	77.1	69.0	59.5	49.7	(31)
		MCDB	48.3	50.7	57.3	66.1	72.0	78.4	85.5	86.2	81.3	72.9	63.3	53.9	(32)
	10%	WB	42.0	44.2	50.3	59.0	65.4	72.3	78.1	79.1	76.1	66.5	57.5	47.5	(33)
		MCDB	47.0	49.3	55.4	63.8	70.5	76.8	84.1	85.1	80.2	71.1	61.6	52.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	15.5	16.9	17.3	16.0	13.0	12.2	13.3	13.3	15.0	15.7	14.9	(35)
		MCDBR	16.9	19.0	20.9	21.1	20.9	17.0	16.0	16.5	15.5	16.9	17.6	17.0	(36)
	5% WB	MCWBR	11.0	12.3	12.7	11.6	10.5	8.1	6.6	6.4	7.0	8.7	10.5	11.1	(37)
		MCWBR	15.1	15.7	16.4	15.6	15.3	12.8	13.5	13.5	11.8	12.8	13.7	14.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.394	0.433	0.494	0.534	0.542	0.565	0.560	0.532	0.491	0.456	0.415	0.392	(40)
		taud	2.169	2.075	1.930	1.882	1.931	1.945	2.011	2.079	2.162	2.176	2.199	2.191	(41)
	Ebn at Noon		247	251	246	243	243	237	237	241	245	242	239	239	(42)
		Edh at Noon	37	45	57	62	60	59	55	51	45	41	36	34	(43)
(44) All-Sky Solar Radiation	RadAvg		746	965	1268	1515	1643	1479	1554	1600	1264	1022	803	672	(44)
	RadStd		50	76	88	148	180	138	214	144	115	104	73	50	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page